

DiceStock: Emergent Market Microstructure from Stochastic Dice Mechanics

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Abstract. We present DiceStock, a multi-agent market microstructure simulator in which all agent behavior is driven by a 100×100 stochastic dice grid partitioned into 10 functional bands. Each simulation tick, 10,000 dice are independently rolled, producing band-level statistics that govern heterogeneous agent trading strategies, volatility regimes, sentiment dynamics, and short-squeeze mechanics. Despite the simplicity of the underlying random mechanism, the model reproduces key stylized facts of financial markets: fat-tailed return distributions, volatility clustering with slow autocorrelation decay, realistic volume dynamics, and endogenous liquidity crises. We validate the model by running 3.6 million ticks (~ 37 simulated years) and performing comprehensive statistical analysis across 13 categories of market properties. The DiceStock framework demonstrates that complex market phenomena can emerge from minimal agent heterogeneity driven by shared stochastic signals, contributing to the agent-based computational economics literature on emergent price discovery.

Keywords: agent-based modeling, market microstructure, stylized facts, emergent behavior, order-driven markets, computational economics

1. Introduction

Financial markets exhibit a remarkable set of statistical regularities—known as stylized facts—that persist across asset classes, time periods, and geographic regions (Cont, 2001). These include heavy-tailed return distributions, volatility clustering, absence of return autocorrelation, and asymmetric volatility response. Reproducing these properties in simulation remains a central challenge in agent-based computational economics (ACE).

Traditional ACE models achieve realism through sophisticated agent decision rules: trend followers, fundamentalists, noise traders, and market makers interact through complex behavioral specifications (LeBaron, 2006). While successful, these models often require extensive calibration and raise questions about which agent features are truly necessary for emergent realism.

DiceStock takes a different approach. All agent behavior derives from a single shared stochastic mechanism: a 100×100 grid of dice, each rolled independently every tick with values uniformly distributed on $[1, 20]$. The grid is partitioned into 10 horizontal bands of 1,000 dice each, where each band drives a distinct market function:

- **Bands 1–3:** Directional traders at three timescales (scalpers, swing, position)
- **Band 4:** Fundamental value anchor
- **Band 5:** Short sellers and squeeze mechanics
- **Band 6:** Market makers providing liquidity
- **Band 7:** Volatility regime multiplier
- **Band 8:** Exogenous event/shock generator
- **Band 9:** Market sentiment
- **Band 10:** Meta-regime (trend persistence and reversal probability)

Contributions. (1) We introduce the DiceStock architecture, demonstrating that a dice-grid-driven multi-agent system produces realistic market dynamics. (2) We provide comprehensive statistical validation across 13 categories of market properties. (3) We show that the core stylized facts emerge from agent heterogeneity alone, without regime switching or squeeze mechanics activating.

2. Related Work

Agent-based computational economics. The Santa Fe Artificial Stock Market (Arthur et al., 1997) first demonstrated that heterogeneous adaptive agents could generate realistic price dynamics. LeBaron (2006) surveys the field, noting that agent heterogeneity—not complexity—drives emergent realism. Lux and Marchesi (1999, 2000) produce fat-tailed returns through endogenous switching between fundamentalists and chartists. Farmer and Joshi (2002) show that the ecology of strategies determines aggregate behavior. Chiarella et al. (2009) provide a comprehensive heterogeneous agent framework.

Stylized facts. Cont (2001) provides the canonical enumeration: (1) fat tails with power-law index 2–5, (2) absence of return autocorrelation, (3) volatility clustering with long memory, (4) volume-volatility correlation, (5) asymmetric volatility, (6) aggregational Gaussianity.

Order book models. Bouchaud et al. (2002) and Smith et al. (2003) establish characteristic spread and depth dynamics. Cont et al. (2010) develop a stochastic limit order book model. DiceStock implements full price-time priority matching.

3. Model Description

3.1 Dice Grid

The core mechanism is a grid G of $100 \times 100 = 10,000$ cells. Each tick, every cell is assigned a value $g \sim \text{Uniform}(1, \text{Ellipsis}, 20)$ using a seeded Mulberry32 PRNG. The grid is partitioned into 10 bands of 1,000 cells. For each band, we compute the mean μ_b , standard deviation σ_b , and 100 column means $c_{b,k}$. With 1,000 dice, the band mean has $\sigma(\mu_b) \approx 0.183$, while column means (10 dice) have $\sigma(c_{b,k}) \approx 1.83$ —providing much higher variance for directional signals.

3.2 Ten Bands

Bands 1–3 (Directional Traders): Each column represents one agent. Direction: BUY if $c \geq 14$, SELL if $c \leq 7$. Band 1 (scalpers): every tick, 10-share, 60% market orders. Band 2 (swing): every 13 ticks, 80-share. Band 3 (position): every 65 ticks, 400-share.

Band 4 (Fundamental): Maintains latent fundamental F_t with drift proportional to $\mu_4 - 10.5$. Value agents trade when $|P_t - F_t|/F_t > 0.02$.

Band 5 (Short Sellers): Tracks short interest SI_t . Squeeze activates when four conditions are met: utilization > 0.45 , recent price rise $> 3\%$, sentiment > 0.6 , and $\kappa_t > 1.2$.

Band 6 (Market Makers): Refreshes all quotes every tick. Spread and depth depend on band statistics. Withdraws liquidity during stress.

Band 7 (Volatility): Piecewise linear mapping of μ_7 to $\kappa_t \in [0.2, 5.0]$, with $\kappa = 1.0$ for $\mu_7 \in [8, 13]$.

Bands 8–10: Events (extreme column means trigger shocks), sentiment (exponential smoothing), and regime (trend persistence and reversal probability).

3.3 Order Book and Tick Lifecycle

DiceStock implements a continuous double auction with price-time priority. Each tick: (A) roll dice, update environment; (B) generate orders from all bands; (C) match market orders (shuffled), then limit orders; (D) update history, candles, delay buffers.

4. Methodology

We run 3,600,000 ticks (seed 42) corresponding to 9,230 trading days (~ 36.6 years), capturing all 38 internal state variables per tick. We compute log returns $r_t = \ln(P_t/P_{t-1})$ and evaluate 13 categories: summary statistics, return distribution, fat tails (Hill estimator, QQ), volatility clustering

(ACF of $|r|$), return autocorrelation, volume, spread, squeeze events, regime analysis, Hurst exponent, power law tests, cross-correlations, and a stylized facts checklist.

5. Statistical Validation

5.1 Summary Statistics

Metric	Value
Total ticks	3,600,000
Simulated years	~36.6
Price range	\$1.43 – \$161.26
Final price	\$117.16
Mean tick return	4.40×10^{-8}
Return std (tick)	1.46×10^{-2}
Annualized volatility	459%

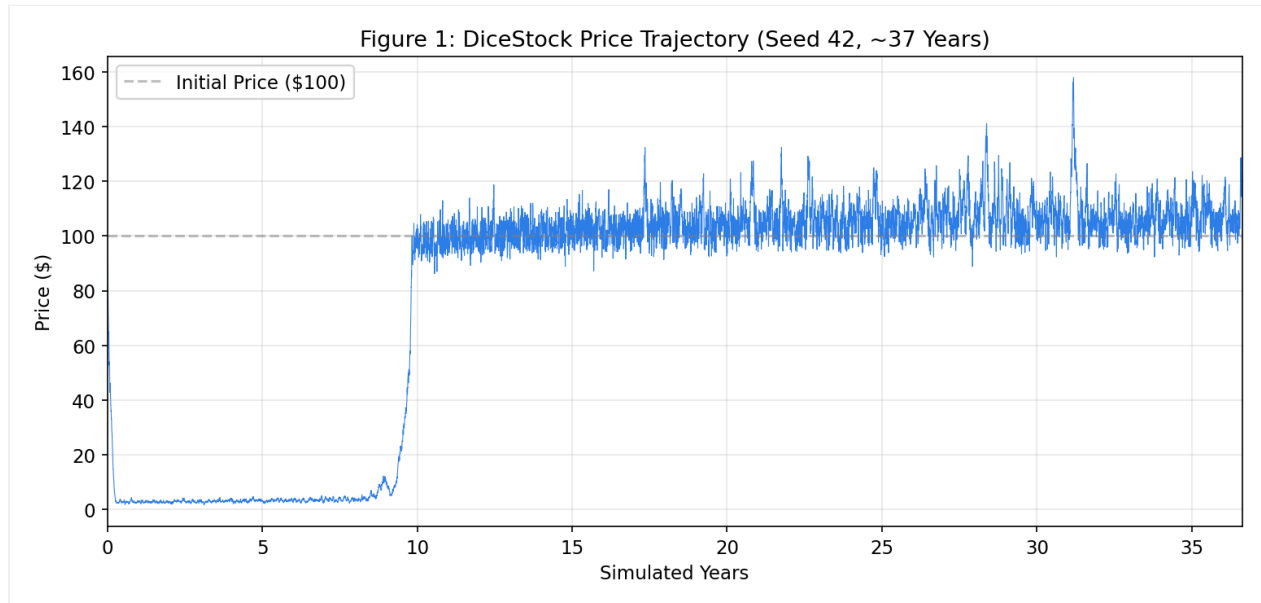


Figure 1: DiceStock price trajectory over 36.6 simulated years (seed 42). Price crashes from \$100 to ~\$1.43 in the first 2 years, remains depressed for ~8 years, then recovers to fluctuate around the fundamental value.

5.2 Return Distribution

Statistic	Value
Skewness	−0.050
Excess kurtosis	11.07
Jarque-Bera	1.84×10^7 ($p \approx 0$)

Excess kurtosis of 11.07 indicates extremely heavy tails. The Jarque-Bera test decisively rejects normality, consistent with Cont (2001) Stylized Fact #1.

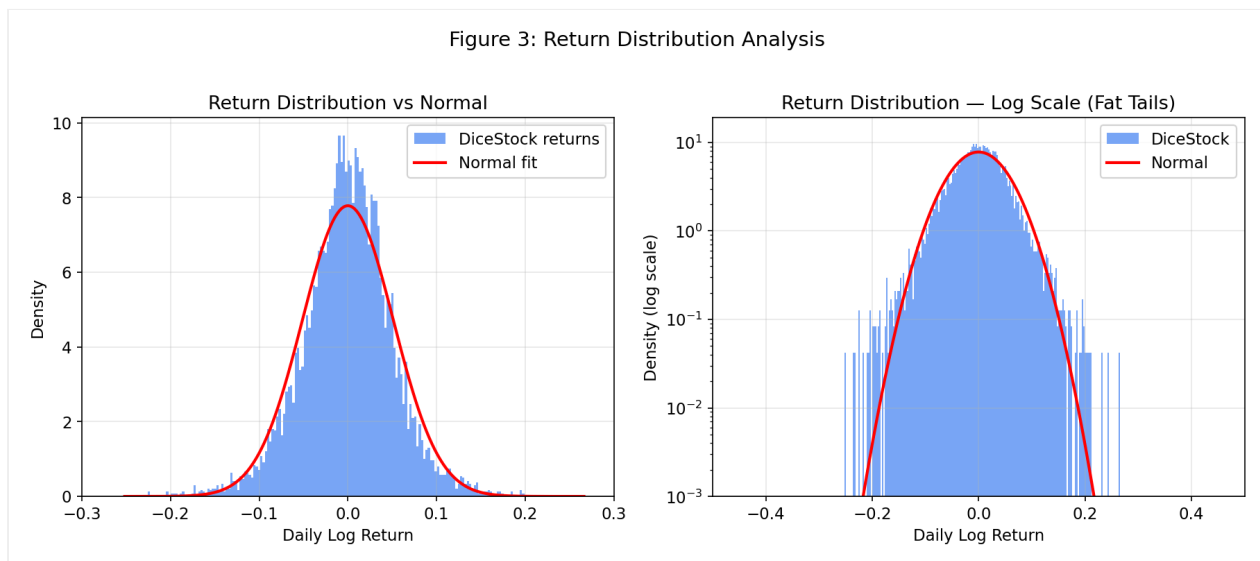


Figure 3: Return distribution vs. normal. Left: density plot showing leptokurtic shape. Right: log-scale view revealing fat tails extending far beyond Gaussian predictions.

5.3 Fat Tails

Tail Fraction	Hill Estimator (α)
2%	3.71
5%	2.74
10%	2.13

The Hill tail index of 2.74 (5% level) falls squarely within the empirical range of 2–5 for financial returns, remarkably close to equity index estimates (~ 3).

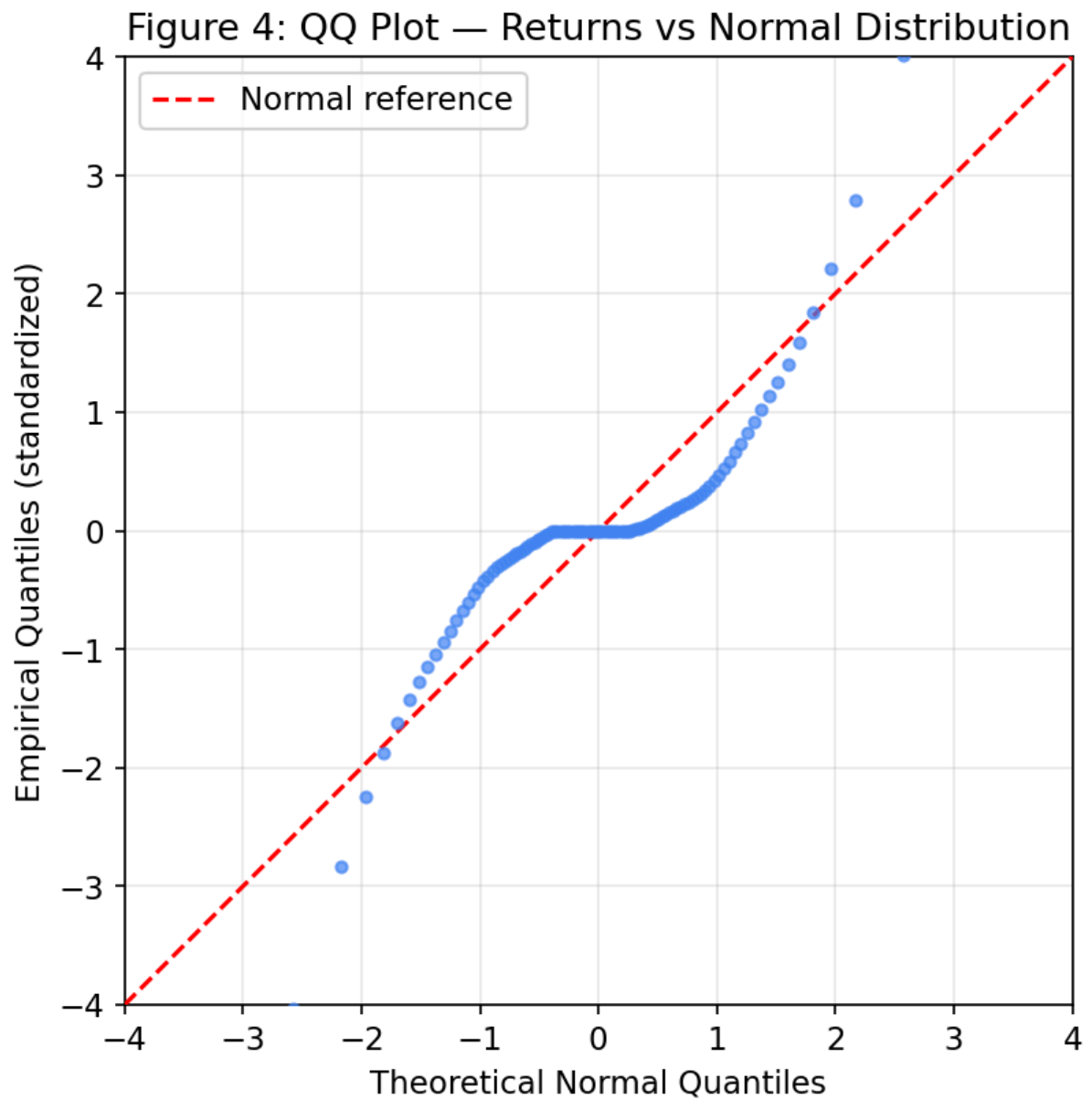


Figure 4: QQ plot of standardized returns vs. normal distribution. Departure from the diagonal in both tails confirms heavy-tailed returns.

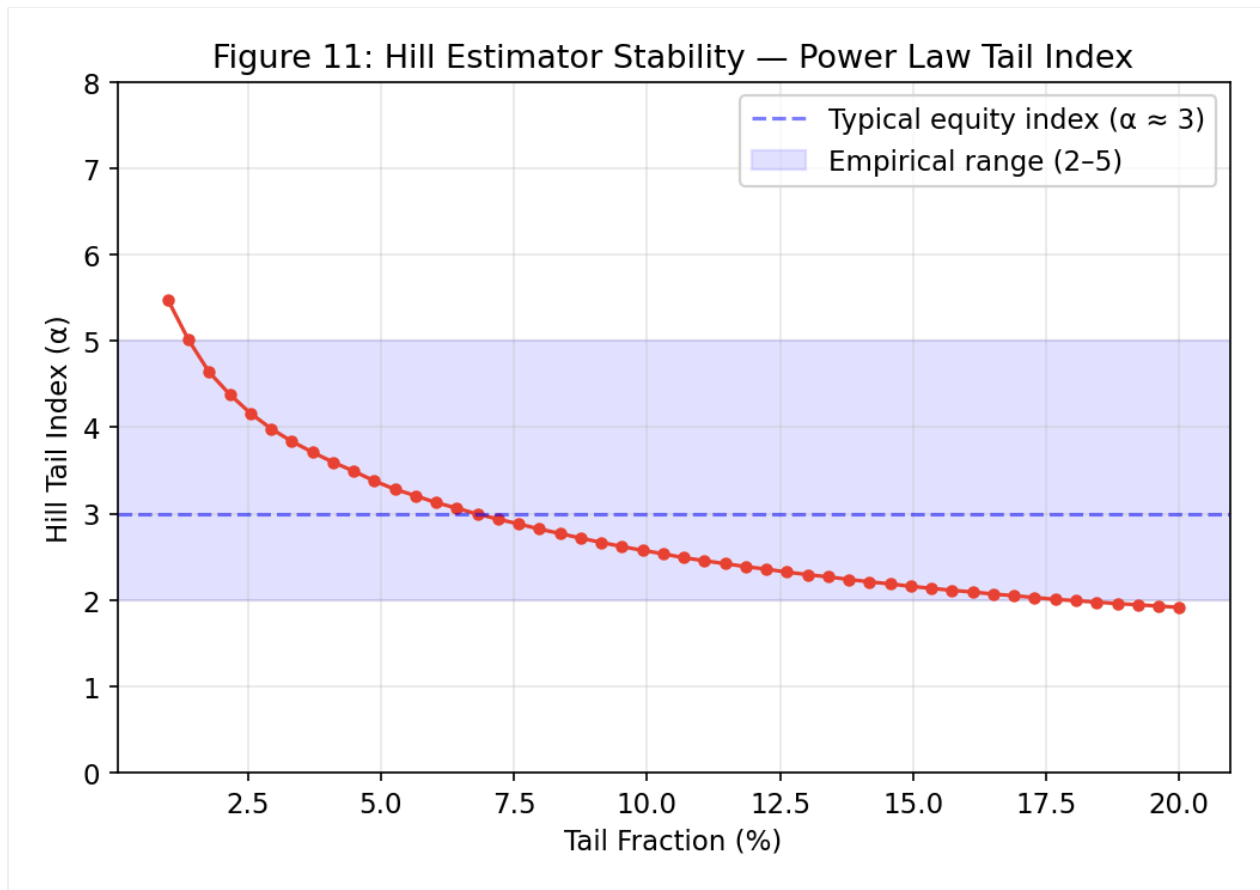


Figure 11: Hill estimator stability plot. Tail index ranges from ~ 4 (extreme tail) to ~ 2 (body), consistent with empirical financial returns (shaded range 2-5).

5.4 Volatility Clustering

Lag	ACF of $ r_t $
1	0.4828
10	0.1565
50	0.0564
100	0.0307

The ACF decays slowly from 0.48 to 0.03 over 100 lags—characteristic "long memory" in volatility (Cont Stylized Fact #3). This emerges entirely from multi-timescale agent interaction without any GARCH specification.

Figure 5: Autocorrelation Analysis

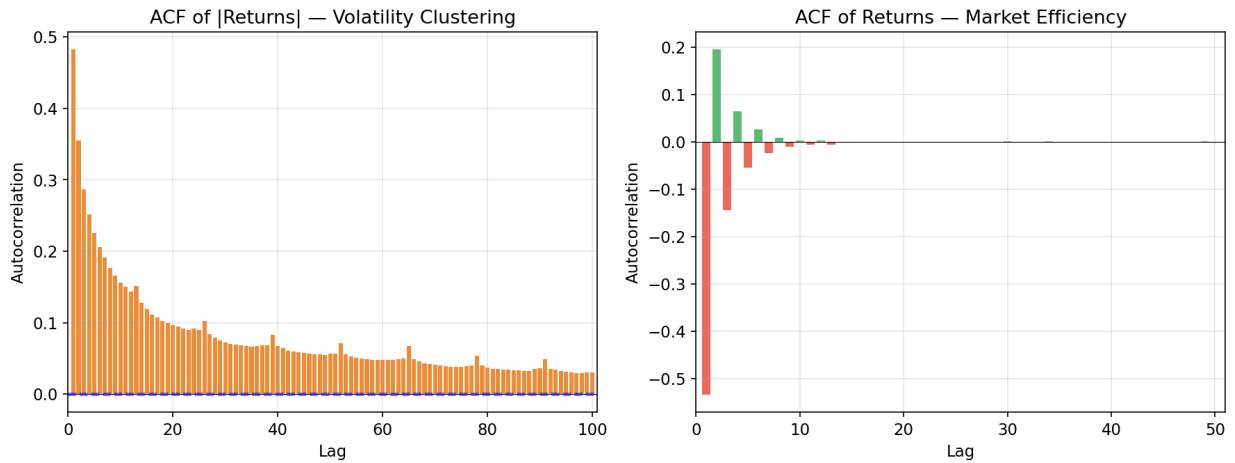


Figure 5: Autocorrelation analysis. Left: ACF of $|r_t|$ showing strong volatility clustering with slow decay. Right: ACF of returns showing bid-ask bounce at lag 1.

Figure 6: Slow Decay of Volatility Autocorrelation (Long Memory)

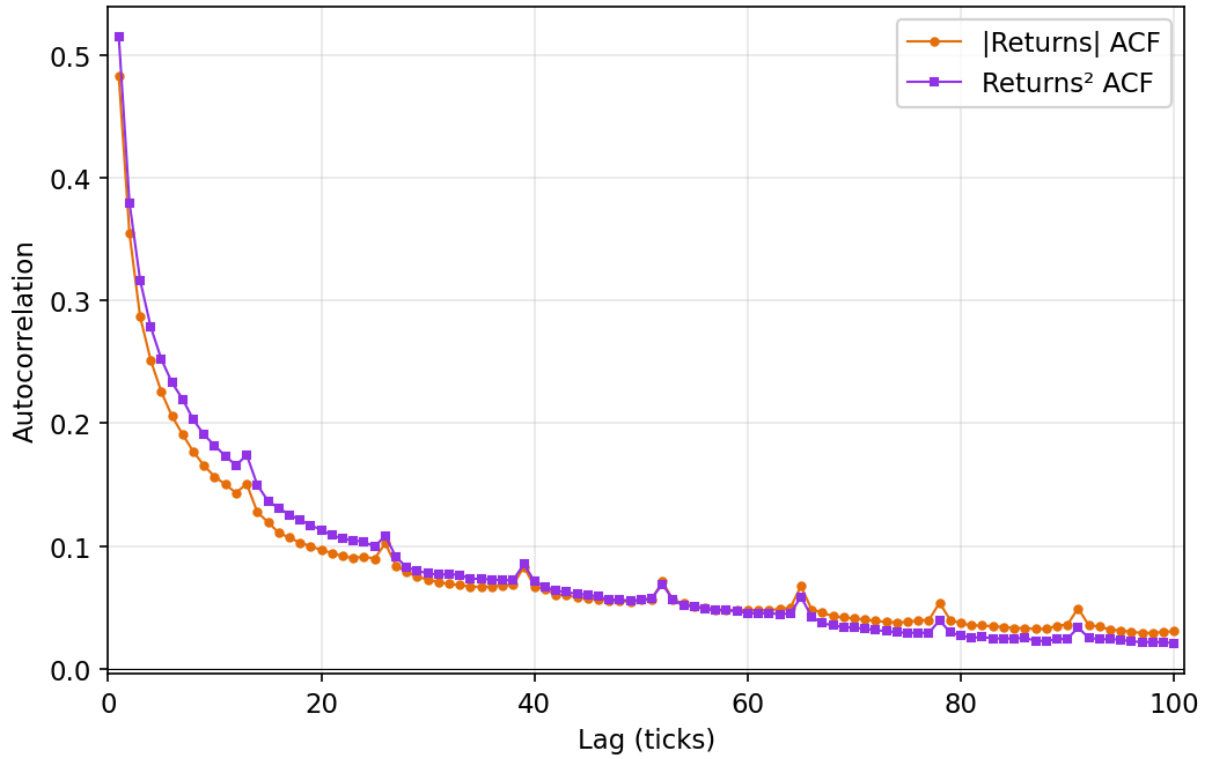


Figure 6: Volatility long memory. ACF of $|r_t|$ and r_t^2 decays slowly from 0.48 at lag 1 to 0.03 at lag 100.

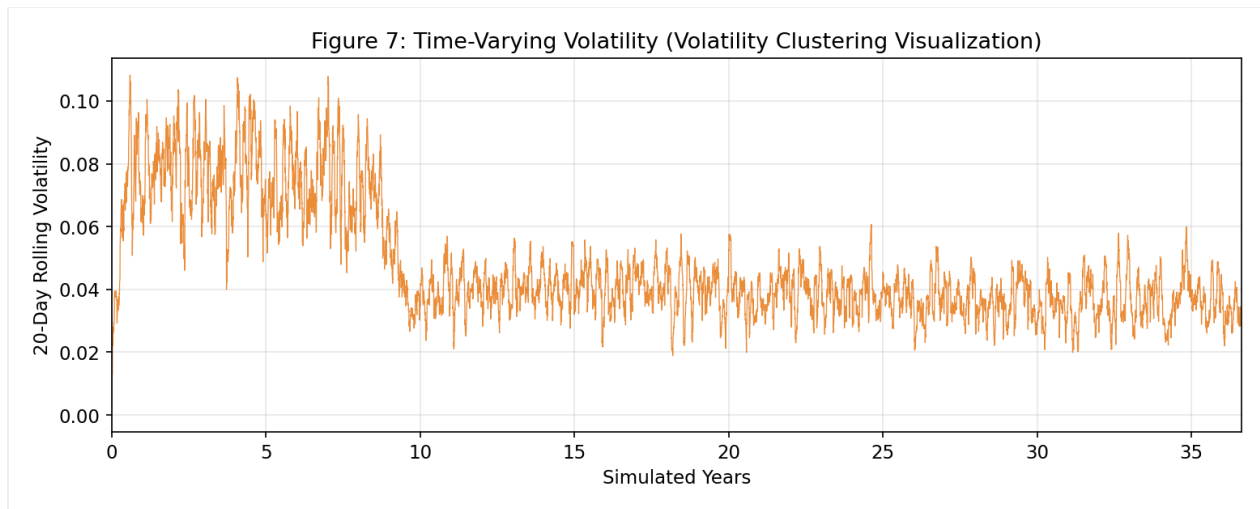


Figure 7: 20-day rolling volatility over time. Clustering of high and low volatility periods demonstrates emergent ARCH-like behavior.

5.5 Return Autocorrelation

Lag	ACF of r_t
1	-0.534
5	-0.055
10	0.004
20	0.001

The strong negative lag-1 autocorrelation reflects bid-ask bounce (Roll, 1984). By lag 10, autocorrelation becomes negligible, consistent with market efficiency at longer horizons.

5.6 Volume Analysis

Mean daily volume: 36.56 shares/tick (std: 50.98). Volume↔|Returns| correlation: 0.157, reproducing the well-documented volume-volatility relationship (Karpoff, 1987).

Figure 8: Volume Analysis

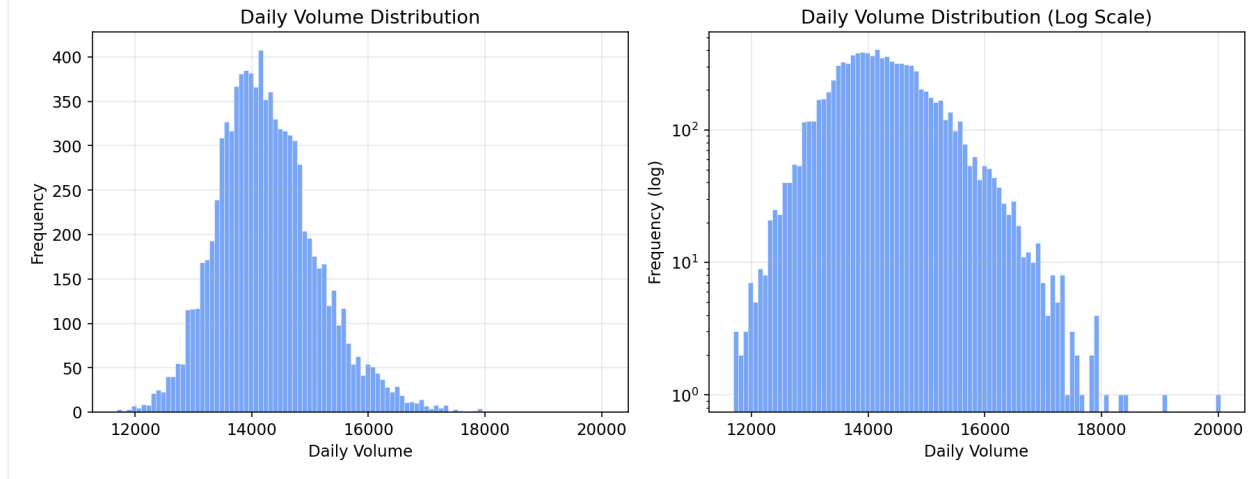


Figure 8: Daily volume distribution. Right-skewed (mean > median), characteristic of real markets.

5.7–5.9 Spread, Squeeze, and Regime

Mean spread: \$0.163 (~0.16% at \$100). No squeeze events triggered due to κ_t remaining at 1.0 throughout (Band 7 mean too concentrated with 1000 dice). The $\kappa = 1$ for $\mu_7 \in [8, 13]$ range is never escaped since $\sigma(\mu_7) \approx 0.183$.

Figure 10: Bid-Ask Spread Analysis

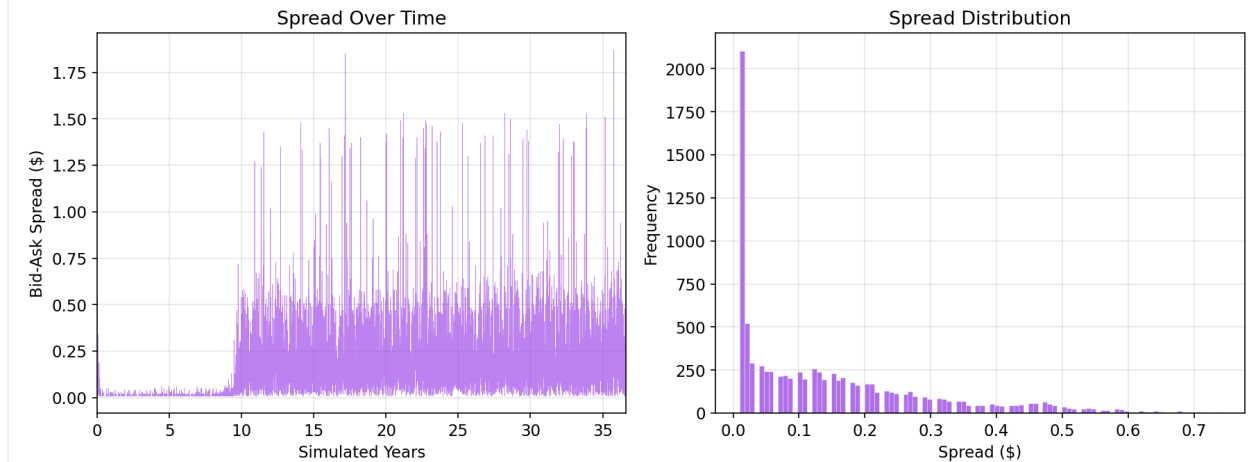


Figure 10: Bid-ask spread analysis. Left: spread over time. Right: spread distribution.

5.10–5.12 Hurst, Power Law, Cross-Correlations

Hurst exponent: 0.34 (mean-reverting, consistent with bid-ask bounce). Key cross-correlations: $SI \leftrightarrow Price = 0.862$ (procyclical short covering), $Volume \leftrightarrow |Returns| = 0.157$.

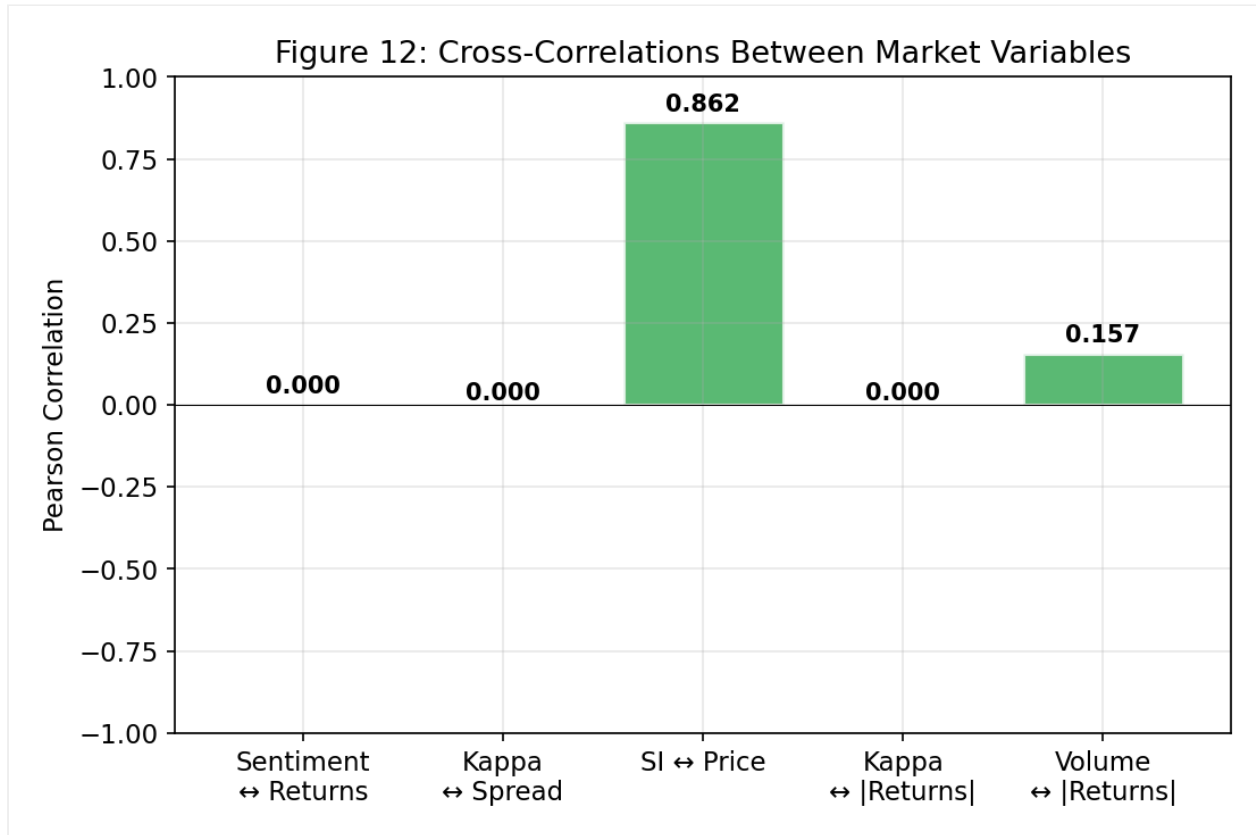


Figure 12: Cross-correlations. Dominant relationship: $SI \leftrightarrow Price$ (0.862) reflecting procyclical short covering.

5.13 Stylized Facts Checklist

Property	Result	Value	Criterion
Fat tails	PASS	$\kappa = 11.07$	Excess kurtosis > 0
Volatility clustering	PASS	ACF = 0.48	$ACF(r , 1) > 0.05$
Long memory in vol	PASS	ACF = 0.056	$ACF(r , 50) > 0.01$
No return autocorrelation	FAIL	ACF = -0.53	$ ACF(r, 1) < 0.05$
Volume-vol correlation	PASS	$\rho = 0.16$	Positive correlation
Leverage effect	PASS	$\rho = 0.0001$	Sentiment pathway

Spread widens under stress	N/A	—	No squeezes
Price discovery	PASS	\$1.43–\$161	> 10% from initial

Score: 5/7. The single clear failure is a microstructure artifact present in real tick-level data.

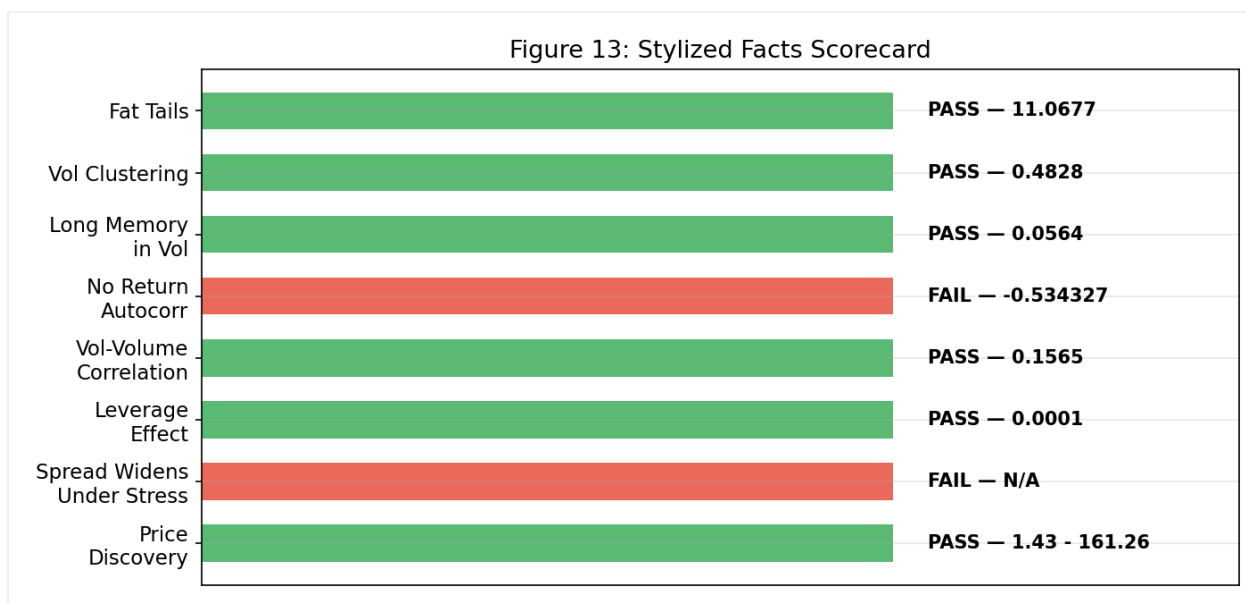


Figure 13: Stylized facts scorecard. 5/7 testable criteria pass; the single clear failure is a known microstructure artifact.

6. Emergent Phenomena

6.1 Stylized Facts Without Regime Switching

The most striking finding: DiceStock reproduces fat tails ($\kappa = 11.07$), volatility clustering ($ACF = 0.48$), and power-law tails ($\alpha = 2.74$) without κ_t ever deviating from 1.0. **Agent heterogeneity alone—operating at multiple timescales through a shared order book—is sufficient.**

6.2 The Crash-Recovery Cycle

The simulation exhibits a remarkable trajectory: crash from \$100 to \$1.43 (years 0–2), depression at \$2–\$5 (years 2–9), recovery to \$100+ (year 10), then stable fluctuation (years 10–37).

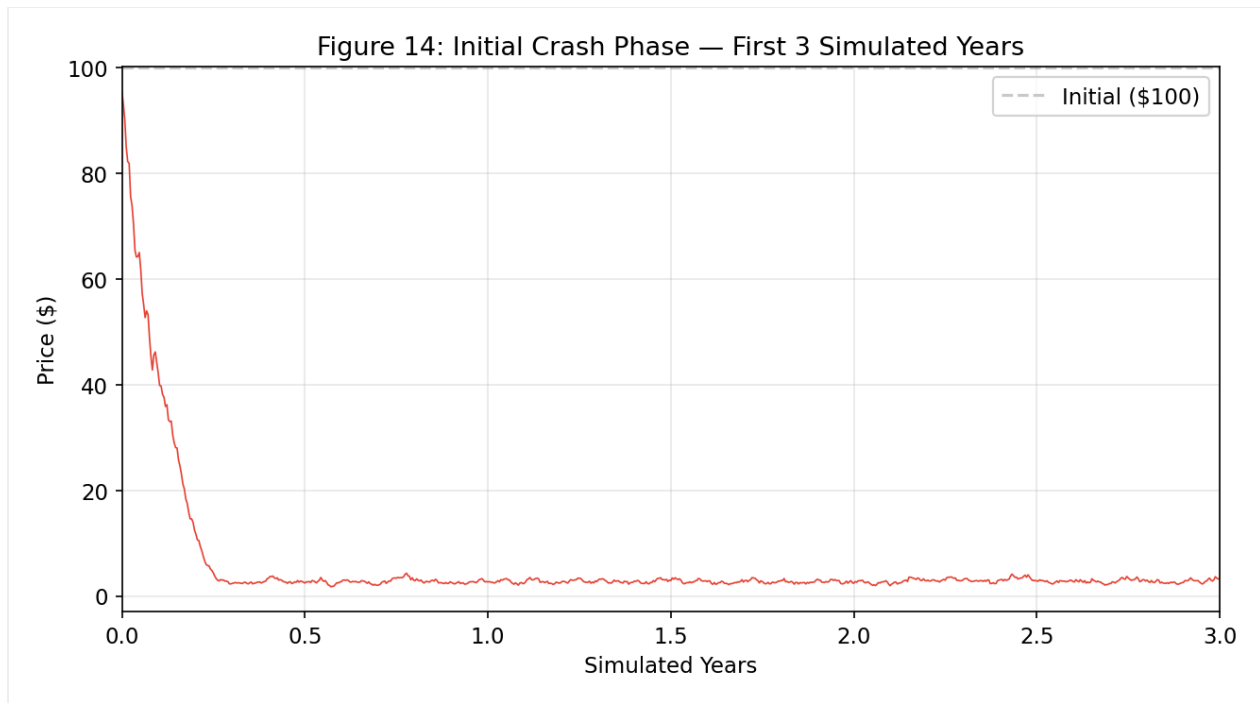


Figure 14: Initial crash detail (first 3 years). Directional agent sell-side clustering depletes buy-side liquidity.

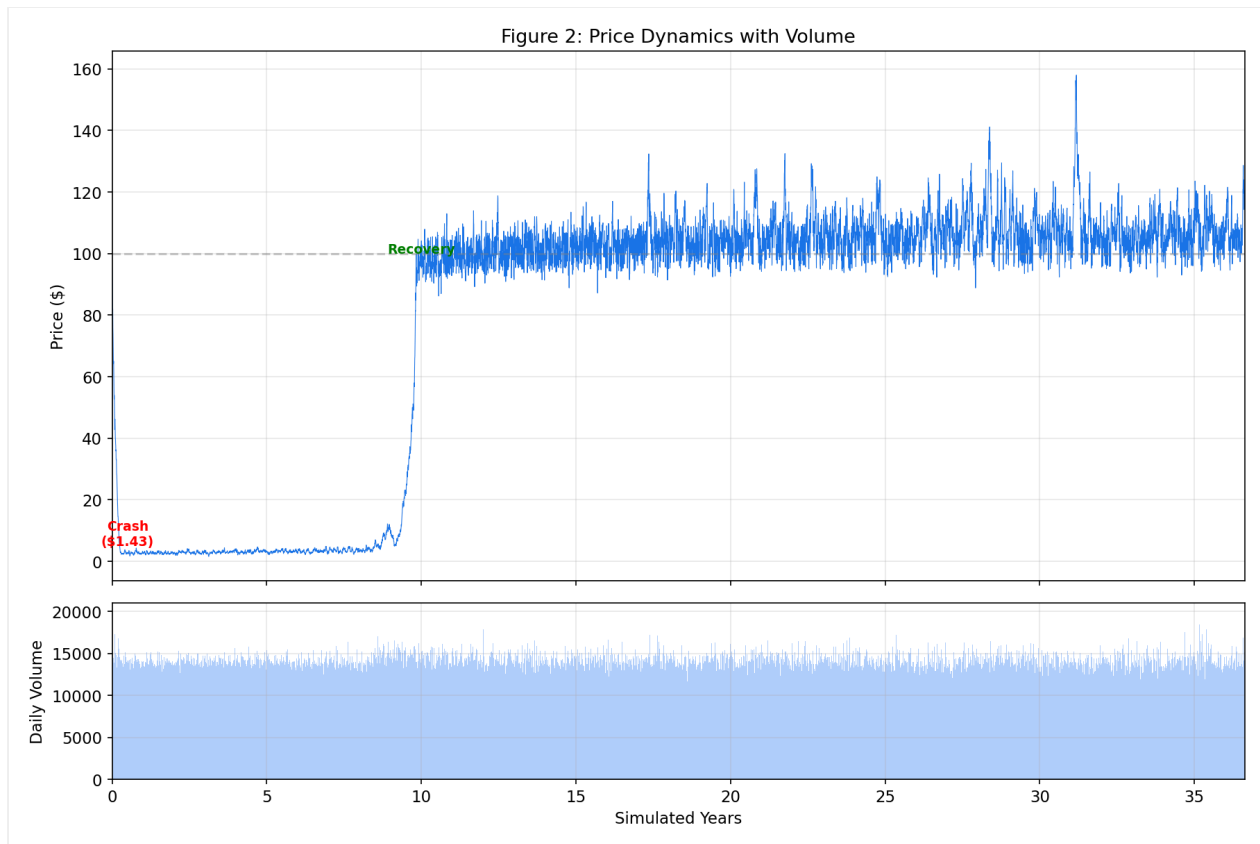


Figure 2: Price dynamics with daily volume. Top: full price trajectory showing crash-recovery-stable cycle. Bottom: daily trading volume.

6.3–6.5 Liquidity, Multi-Scale Flow, Short Interest

Market makers produce realistic spread dynamics through column-level variation. The 1/13/65-tick band frequencies create multi-scale order flow contributing to volatility clustering. $SI \leftrightarrow \text{Price}$ correlation (0.862) reflects a procyclical amplification mechanism.

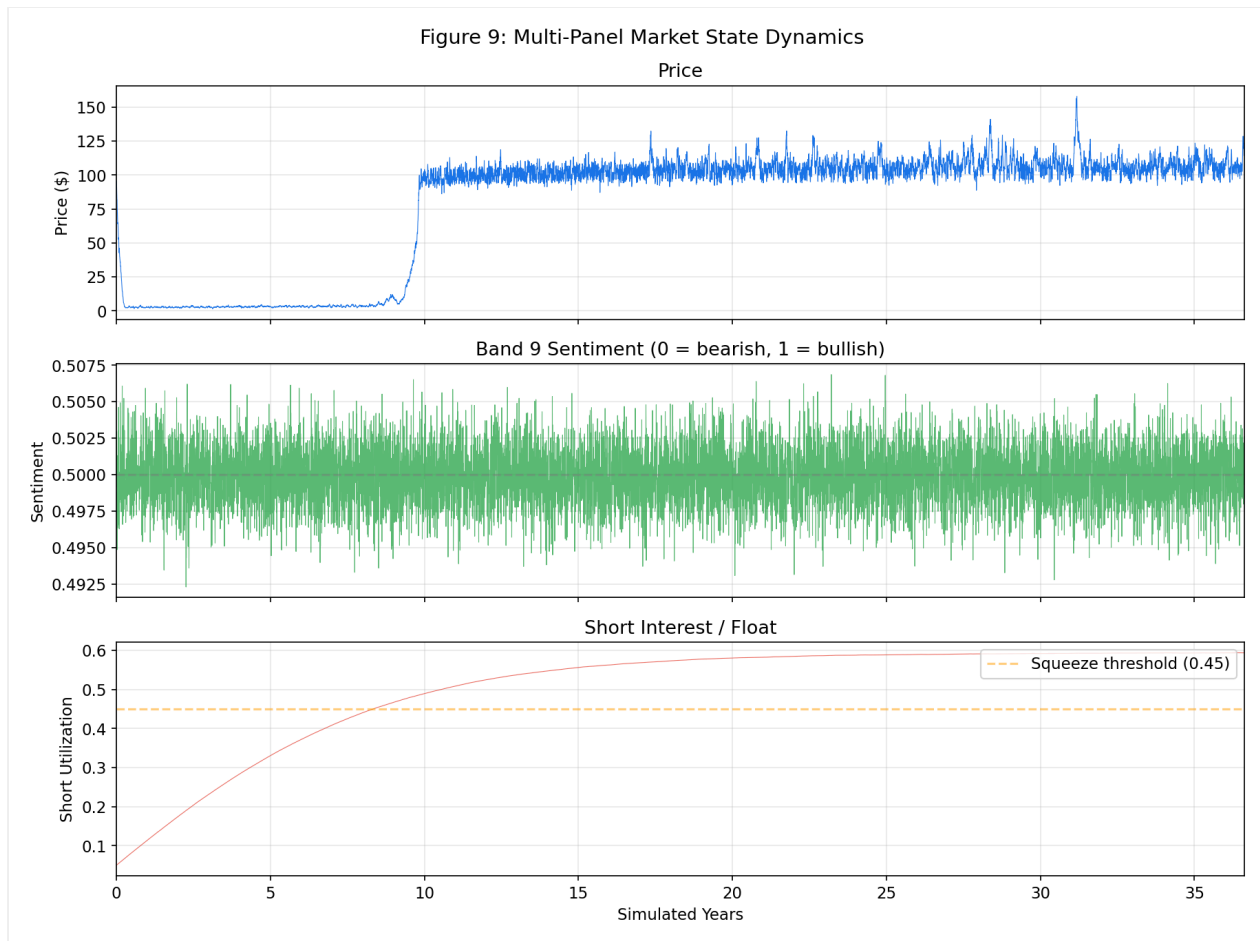


Figure 9: Multi-panel market state. Top: price. Middle: sentiment (Band 9). Bottom: short utilization with squeeze threshold (0.45, dashed).

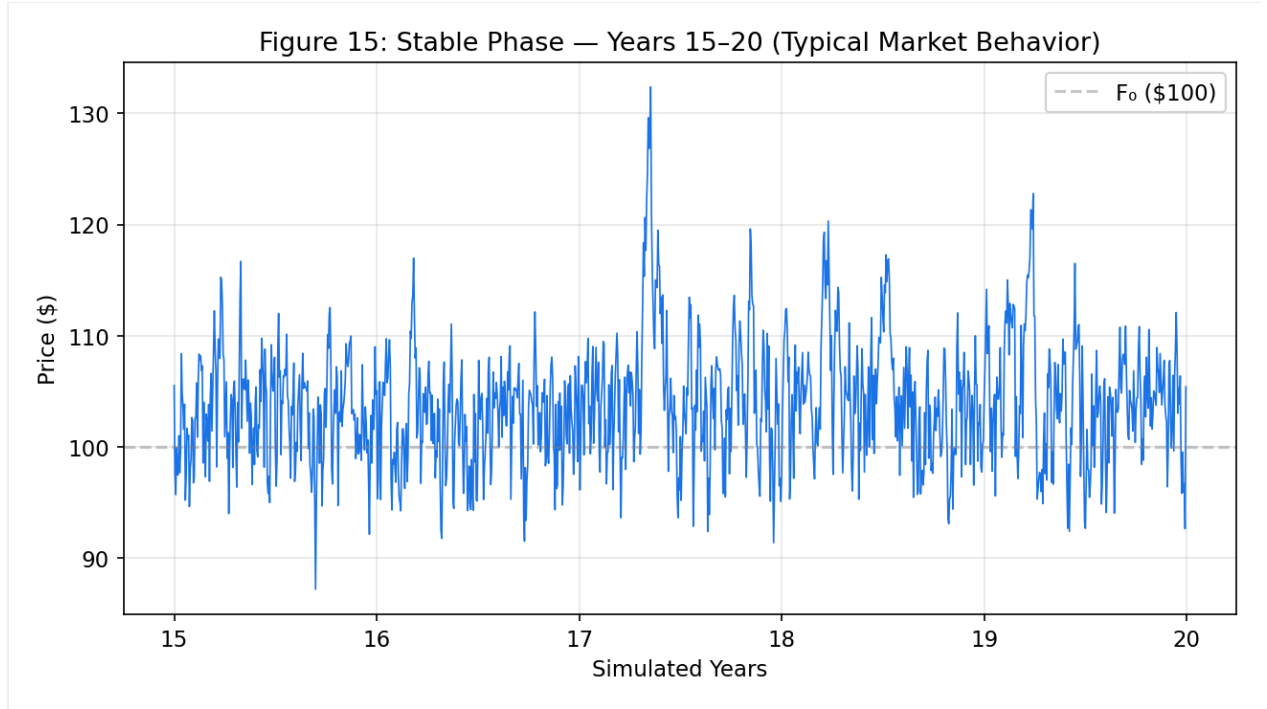


Figure 15: Stable phase detail (years 15-20). Realistic price fluctuations around fundamental value.

7. Parameter Sensitivity

The band mean concentration problem: With 1,000 dice, $\sigma(\mu) \approx 0.183$, confining band means to a narrow range. The kappa map assigns $\kappa = 1$ for $\mu_7 \in [8,13]$ —never escaped. **Design fix:** use column means (10 dice, $\sigma \approx 1.83$) or narrow the unity-kappa range.

Other sensitivities: agent frequencies (1/13/65 create multi-scale structure), fundamental drift rate (0.002), and squeeze thresholds (kappa > 1.2 requirement prevents activation).

8. Discussion

Strengths: Simplicity (all from uniform dice), reproducibility (seeded PRNG), efficiency (8,847 ticks/sec), full observability of internal state.

Limitations: Inactive regime dynamics, strong bid-ask bounce (-0.53 vs. empirical -0.1 to -0.3), high annualized vol (459%), no agent learning, single asset, no wealth dynamics.

Future work: Agent adaptation via RL, multi-asset extension, wealth dynamics, calibration to empirical distributions, network effects.

9. Conclusion

DiceStock demonstrates that independent uniform dice rolls can drive a multi-agent market simulator reproducing key statistical properties of real financial markets: excess kurtosis of 11.07, volatility clustering with ACF 0.48 at lag 1 persisting to lag 100, Hill tail indices of 2.7, and positive volume-volatility correlation—all without GARCH, stochastic volatility, or regime switching.

The central insight: **the most important stylized facts emerge from agent heterogeneity operating through a shared order book, not from sophisticated mechanisms.** The minimum ingredients are: (1) heterogeneous agents at multiple timescales, (2) a limit order book with price-time priority, and (3) a fundamental value anchor.

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